Yuning Wu

ML+Robotics Researcher / Creator / Maker / Designer
yuningw@andrew.cmu.edu • 412-626-2495 • https://ice-5.github.io/

With an interdisciplinary background in machine learning, robotics, and computational design, I'm constantly motivated by the vision of AI helping us build a better cyber-future, both virtually and concretely. I enjoy prototyping and making things from 0 to 1. I enjoy exploring solutions to critical problems. I'm a do-er.
Machine Learning, Robotics, Human-Robot Interaction & Interface, Computational Design

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EDUCATION /

Carnegie Mellon University. Pittsburgh, USA.
Ph.D. Computational Design — 2019-Present
Co-Advisors: Daniel Cardoso Llach (Computational Design), Jean Oh (Robotics)
My research focuses on prototyping robots that can work collaboratively with humans in repetitive work duties and sites, such as building our living space

M.S. Machine Learning — 2019-2021
Research Advisor: Katerina Fragkiadaki (Machine Learning)

Tsinghua University. Beijing, China. — 2010-2017
B.Arch. M.Arch. Architecture
With a focus on generative design, 3D prototyping, and digital fabrication

WORK /

Research Intern
Autodesk Research, AI Lab & Robotics Lab. Remote, USA. — 2021
Collaborate with Jieliang Luo (Principal AI Scientist), Hui Li (Sr. Research Scientist)
• Work on inverse reinforcement learning (IRL) in contact-rich robotic manipulation

Co-Founder, CEO
Metascope Technology. Beijing, China. — 2017-2019
• Build an aesthetics recommendation system with ML and computer vision
• Profile clients' preferences of spatial aesthetics
• Use statistical results to help architects produce better designs

Architect
Nikken Sekkei. Tokyo, Japan. — 2014
• Practice architecture in two major urban design and high-rise projects
• Investigate how 3D generative design can boost the interior design of buildings

RESEARCH & PROJECT /

Project Tech Lead
Rethinking Automation in Construction: Towards a Robotically-Supported Collaborative Construction (Manufacturing PA Program) — 2021-Present
Advised by Daniel Cardoso Llach, Jean Oh
• Prototype collaborative robots creating/constructing 3D space along with humans
• Apply reinforcement learning (RL) based social-navigation algorithms
• Develop pedestrian tracking and detection systems with 2D Lidar and camera
• Develop gesture recognition in noisy environments for human-robot interaction

Principal Researcher
Teaching AI Spatial Sense and Aesthetics by Playing LEGOs — 2021-Present
Advised by Katerina Fragkiadaki
• Build a GNN-based physics engine to infer the stability of a stacked LEGO structure
• Investigate hierarchical reinforcement learning (HRL) for sequence pattern extraction

Project Tech Lead

Semi-Autonomous Construction in Dynamic Environments: A Reinforcement Learning Approach Towards Scalable and Robust Multi-Agent Drone Construction 2020

Advised by Daniel Cardoso Llach, [video, code]

• Raise a semi-autonomous RL framework for dynamic solutions to real-world drone construction applications

• Develop and apply decentralized, RL-based multi-agent collision avoidance algorithm in building flexible 3D forms

Learning Dense Reward for Partially Observable Contact-Rich Manipulation 2021

Advised by Jieliang Luo, Hui Li, [code]

• Use self-supervised learning to train an end-to-end dense reward system

• Consider partially observable, contact-rich manipulation tasks as a scene

• Raise a data-efficient sampling method for learning from demonstrations (LoD)

System Design Team Member

MoonRanger 2020

Advised by Heather Jones, William (Red) Whittaker, [project]

• MoonRanger is a rover designed to explore the lunar surface searching for water.

• It is scheduled for launch in 2022 via collaborative effort from CMU, NASA, and Astrobotics

PAPER / IN PREPARATION

Y. Wu, Jieliang Luo, Hui Li. **Learning Contact-Rich Manipulation by Geometrically-Informed Trial-and-Error.** To be submitted to *International Conference on Robotics and Automation*, ICRA 2022.

PUBLISHED

Z. Fang, **Y. Wu**, A. Hassonjee, A. Bidgoli, D. Llach. **Towards an Architectural Framework for Distributed, Robotically-Assisted Construction: Using Reinforcement Learning to Support Scalable Multi-Drone Construction in Dynamic Environments**. In *Proceedings of the 40th Annual Conference of the Association for Computer-Aided Design in Architecture*, ACADIA 2020. [PDF]

O. Akin*, **Y. Wu***. **State Space Paradox of Computational Research in Creativity**. In *Proceedings of the 19th Heron Island Conference on Computational and Cognitive Models of Creative Design*, HI 2019.

TECHNICAL SKILL /

Programming. Python, C#, C, C++

Machine Learning. PyTorch, Tensorflow+Keras, Spark, Numpy, Scipy, Matplotlib

Simulation & Framework. OpenAI Gym, PyBullet, Unity+MLAgents

3D Modeling and Prototyping. AutoCAD, Rhino+Grasshopper, Revit+Dynamo, CATIA

Robotics. ROS, Unity

COURSE /

Introduction to Machine Learning (Ph.D.) • Advanced Machine Learning • Computer Vision • Machine Learning with Large Dataset • Convex Optimization • Probabilistic Graphical Models • Deep Reinforcement Learning and Control • Space Robotics

HONOR & AWARD /

Autodesk Merit Scholarship / Autodesk Research, Carnegie Mellon University 2021

Manufacturing PA (MPA) Fellowship / State of PA, Carnegie Mellon University 2021

GSA-Provost GuSH Grant / Carnegie Mellon University 2020

Lego Scholarship / Lego Foundation, Carnegie Mellon University 2019

Distinctive Graduation Thesis / Tsinghua University 2017

Audi China Student Formula Racing Commercial Challenge / Tsinghua University 2013

Tsinghua Siyuan Excellent Student Fellowship / Tsinghua University 2010-2014